

OIL INDIA LIMITED - eRTMAC OPERATIONS

OFFICIAL WELL COMPLETION REPORT (WCR) & DRILLING HISTORY
WELL IDENTIFIER: 31/2-21 S

1. Well Location & Technical Header

Operator: Oil India Limited / Joint North Sea	Well Name: 31/2-21 S
UTM Easting (X): 525098.8 m	UTM Northing (Y): 6760005.9 m
Latitude: 60.974231° N	Longitude: 3.463683° E
Total Depth (MD): 3191.6 m	Spud Depth: 1999.9 m
Drilling Rig: eRTMAC Semi-Submersible Rig-07	Status: Plugged & Abandoned (P&A;) / Offset Reference

2. Stratigraphic Progression & Formation Tops

Formation / Stratum Name	Top Depth MD (m)	Lithology Characterization	Drillability / Risk Profile
Vaale Fm.	1999.9 m	Interbedded Sand / Shale	Loss / Depletion
Jorsalfare Fm.	2099.2 m	Interbedded Sand / Shale	Loss / Depletion
Draupne Fm.	2198.5 m	Interbedded Sand / Shale	Loss / Depletion
Heather Fm.	2297.8 m	Interbedded Sand / Shale	Loss / Depletion
Sognefjord Fm.	2397.1 m	Interbedded Sand / Shale	Loss / Depletion
Fensfjord Fm.	2496.4 m	Interbedded Sand / Shale	Loss / Depletion
Ness Fm.	2595.8 m	Interbedded Sand / Shale	Loss / Depletion
Etive Fm.	2695.1 m	Interbedded Sand / Shale	Loss / Depletion
Rannoch Fm.	2794.4 m	Interbedded Sand / Shale	Loss / Depletion
Oseberg Fm.	2893.7 m	Interbedded Sand / Shale	Loss / Depletion
Drake Fm.	2993.0 m	Interbedded Sand / Shale	Loss / Depletion
Cook Fm.	3092.3 m	Interbedded Sand / Shale	Loss / Depletion

3. Casing Design & Mud Program Summary

36 in Hole	30 in Conductor	350 m	Spud Mud / Seawater	1.05 SG	FIT 1.25 SG
26 in Hole	20 in Surface	1,050 m	KCl / Polymer	1.12 SG	LOT 1.45 SG
17-1/2 in Hole	13-3/8 in Interm.	2,200 m	Low Solids Non-Dispersed	1.24 SG	LOT 1.62 SG
12-1/4 in Hole	9-5/8 in Prod.	2892 m	Oil Based / Synthetic Mud	1.32 SG	LOT 1.78 SG
8-1/2 in Hole	7 in Liner	3192 m	Synthetic Oil Based (SOBM)	1.38 SG	LOT 1.88 SG

4. Historical Operational Incidents, NPT & Mitigations (Well: 31/2-21 S)

This section constitutes the institutional memory log extracted by the NWIS knowledge engine. All alerts generated during active drilling cite these exact records.

INCIDENT #1: Lost Circulation / Mud Loss			Severity: HIGH	NPT Hours: 24.0 hrs
Depth Interval: 2150.0 m - 2210.0 m		Formation: SOGNEFJORD FM		Status: Resolved on Site
Event Occurrence Narrative: Complete loss of returns while penetrating unconsolidated Sognefjord oil sand at 2185m. Dynamic losses > 100 bbl/hr.				
Mitigation Action Applied: Pumped cross-linked polymer gunk pill (40 bbls) followed by 50 bbl heavy calcium carbonate squeeze pill. Reduced drill fluid density from 1.25 SG to 1.15 SG.				
Offset Drilling Recommendation & Institutional Lesson: Depleted reservoir pressure in Sognefjord sandstone requires strict equivalent mud weight management under 1.16 SG.				

INCIDENT #2: Differential Sticking			Severity: HIGH	NPT Hours: 18.0 hrs
Depth Interval: 2680.0 m - 2720.0 m		Formation: FENSFJORD FM		Status: Resolved on Site
Event Occurrence Narrative: Differential sticking while taking directional survey at 2698m. Drill collars embedded against permeable filter cake.				
Mitigation Action Applied: Pumped 50 bbl diesel-glycol release pill. Reduced hydrostatic pressure by lightened fluid displacement. String pulled free at 80 klbs overpull.				
Offset Drilling Recommendation & Institutional Lesson: Utilize spiral drill collars and keep string in continuous rotation when stationary on bottom.				